

DATA ANALYTICS CAPSTONE · FINAL PRESENTATION

# Forecasting Short-Term Exchange Rate Movements and Volatility in Major Currency Pairs Using Macroeconomic Indicators and Machine Learning

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Kumud Naithani · Group 8

*Mentor: Abhay Poddar*

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# Executive Summary

- **Objective & Scope.** Compare macroeconomic, machine-learning, sentiment, and policy-event approaches to forecasting short-term direction and volatility for USD/INR, EUR/USD, GBP/USD, and USD/JPY.
- **Why it matters.** The FX market moves USD 7T+ daily; treasuries, traders, and importers/exporters need reliable short-term signals to hedge risk, yet the “Meese-Rogoff puzzle” shows this is genuinely hard.
- **Data.** 2,143 real daily observations per pair (2018-2026) from FRED, plus verified macro series (interest rates, inflation) across 4 economies — 8,572 pooled currency-day records.
- **Key results.** RQ1 full 4-predictor model:  $R^2$  reaches 12.4% for EUR/USD (3 of 4 coefficients significant), while USD/INR, GBP/USD, and USD/JPY stay non-significant ( $R^2$  3.5-5.0%). RQ2: walk-forward GARCH and Random Forest both beat a naive baseline by ~22-25%. RQ4: no significant average volatility spike, but a surprise-magnitude split shows the expected directional pattern.
- **Usage.** Findings feed a practical early-warning dashboard for treasury teams, flagging elevated volatility risk ahead of scheduled policy announcements.

# Gap Analysis

## What the literature establishes

- Meese & Rogoff (1983): structural macro models fail to beat a random walk short-term — the “Meese-Rogoff puzzle” that still motivates this field.
- Separate literatures address each piece: econometric volatility (Engle 1982; Bollerslev 1986), ML forecasting (Breiman 2001; Chen & Guestrin 2016; Hochreiter & Schmidhuber 1997), sentiment (Tetlock 2007; Hutto & Gilbert 2014; Araci 2019), and policy surprises (Kuttner 2001).

### The Gap

No study evaluates macro-fundamental, ML, sentiment, and policy-event approaches together, on the same currency pairs — including an emerging-market pair (USD/INR) alongside major developed pairs — leaving practitioners without comparative evidence on which model family actually helps.

# Research Questions & Hypotheses

## RQ1 Do macro indicators (rate & inflation differentials, trade balance, GDP growth) predict short-term FX direction?

**H0:** None of the macro predictors is significantly associated with directional movement.

**Ha:** At least one predictor is significantly associated with directional movement.

**Method:** Full 4-predictor OLS + Random Forest robustness check, VIF-screened, 4 pairs

● **Result:** Mixed. Reject H0 for EUR/USD ( $R^2=12.4\%$ , 3 predictors sig.); fail to reject for the other three pairs ( $R^2$  3.5-5.0%).

## RQ2 Do ML models (Random Forest) forecast volatility significantly better than GARCH(1,1)?

**H0:** ML forecast error is not significantly different from the GARCH baseline.

**Ha:** ML achieves significantly lower forecast error than GARCH.

**Method:** GARCH(1,1) vs. Random Forest on lagged returns, walk-forward re-estimation, paired-loss t-test

● **Result:** Fail to reject H0 ( $p = 0.78$ ). Walk-forward GARCH RMSE 0.231 vs. RF 0.227 — both beat the naive baseline (0.298) by ~22-25%.

*RQ1 is no longer a clean null — EUR/USD rejects H0 while the other three pairs don't. RQ2 still fails to reject, but cuts real forecast error by ~22-25% vs. naive (see Results).*

# Research Questions & Hypotheses

**RQ3**

**Does financial-news sentiment (GDELT/VADER/FinBERT) improve directional accuracy beyond macro indicators?**

**H0:** Adding sentiment does not significantly improve directional accuracy.

**Ha:** Adding sentiment improves directional accuracy by  $\geq 5$  points.

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**Method:** VADER + FinBERT sentiment index added to best RQ2 model, McNemar's test (planned)

● **Result:** In progress. Architecture defined; GDELT collection is the critical remaining data-engineering step.

**RQ4**

**Do FOMC policy announcements cause a significant spike in USD/INR volatility?**

**H0:** Post-announcement volatility is not significantly different from the pre-announcement baseline.

**Ha:** Post-announcement volatility is significantly higher.

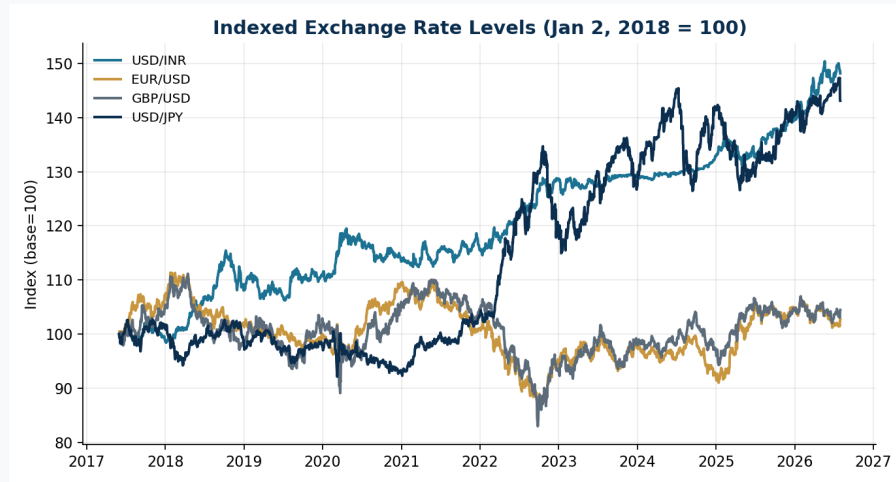
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**Method:** Event study: 71 real FOMC dates (2018-2026), paired t-test on |return|, days -10..-6 vs. 0..+1; split by |3-month T-bill yield change| as a surprise-size proxy

● **Result:** Fail to reject H0 overall ( $p=.84$ ). High-surprise half: +2.2% post-jump; low-surprise half: -6.3% — right direction, underpowered at  $n=36/35$ .

*RQ3 is the one open thread (sentiment pipeline in progress); RQ4's surprise-size split (see Results) points the right way but needs more data to confirm.*

# Data Description & EDA



Indexed FX levels, 4 pairs, 2018-2026 (source: FRED)

**8,572**  
pooled currency-day observations

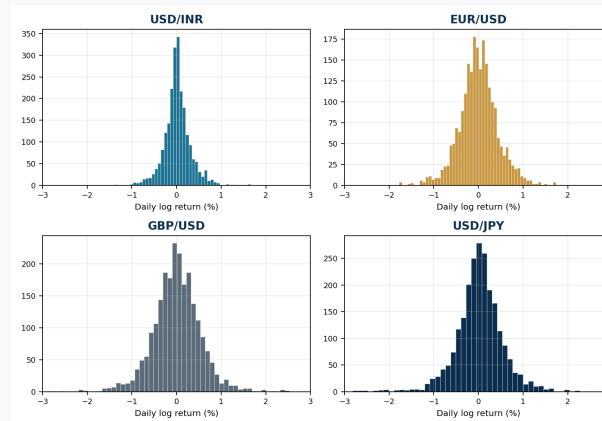
**2018-26**  
8+ years, all non-Kaggle sources

**5.0-8.9%**  
annualized volatility range

**4.3%**  
missing data (structural)

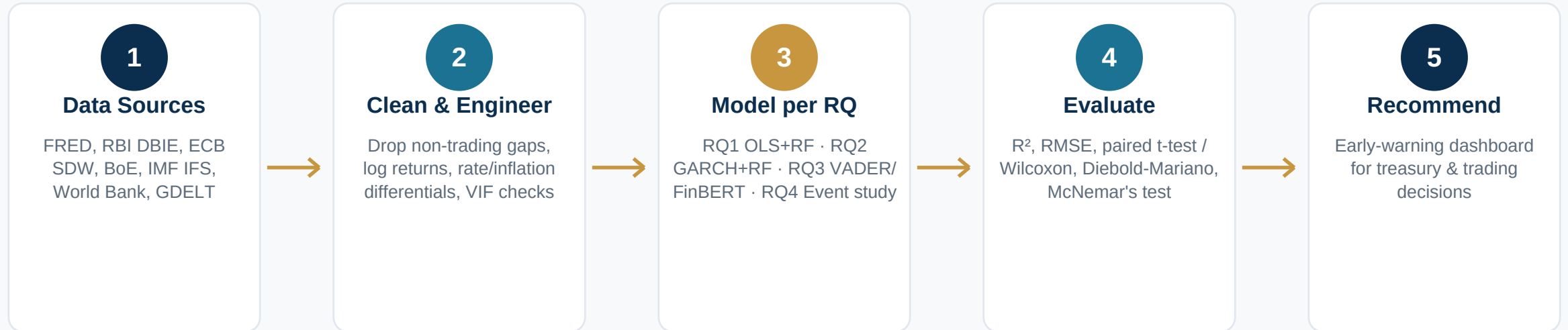
## EDA headline:

All four pairs show excess kurtosis (fat tails), motivating GARCH over constant-variance models for RQ2.



Daily log-return distributions, all 4 pairs — fat tails visible in every panel

# Architecture Diagram / Workflow



**Legend:** Sequential pipeline, left to right. Each stage's output feeds the next; RQ1-RQ4 run in parallel within Stage 3 using a shared, cleaned data foundation from Stage 2.

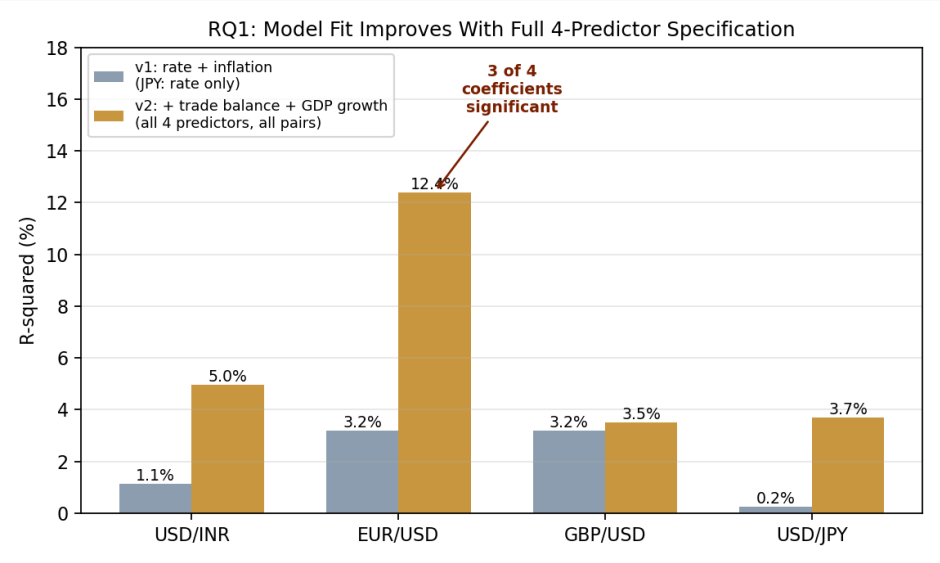
# Model Building

| RQ  | Baseline / Comparator                            | Justification                                                                                                                   |
|-----|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| RQ1 | OLS (linear) vs. Random Forest (nonlinear check) | Tests the linear UIP/PPP relationship directly; RF checks for nonlinearity theory may miss. VIF-screened for multicollinearity. |
| RQ2 | GARCH(1,1) vs. Random Forest (5 lagged returns)  | GARCH is the finance-standard volatility-clustering model; RF tests whether nonlinear ML adds value over it.                    |
| RQ3 | Best RQ2 model vs. + VADER/FinBERT sentiment     | Isolates the incremental value of news sentiment, holding the base model fixed (planned).                                       |
| RQ4 | Pre- vs. post-announcement paired comparison     | Standard event-study design; mirrors Kuttner (2001)'s framework for policy-surprise effects.                                    |

**Rigor built into every model:**

- 5-fold CV (RQ1) and time-series CV with no shuffling (RQ2) to prevent look-ahead leakage
- 80/20 train-test split; GARCH walk-forward refitting (no look-ahead) rather than a single in-sample fit
- RMSE, R-squared, and paired-loss t-tests for honest comparison; VIF < 10 threshold for RQ1 predictors

# Results: RQ1 Multi-Predictor Model



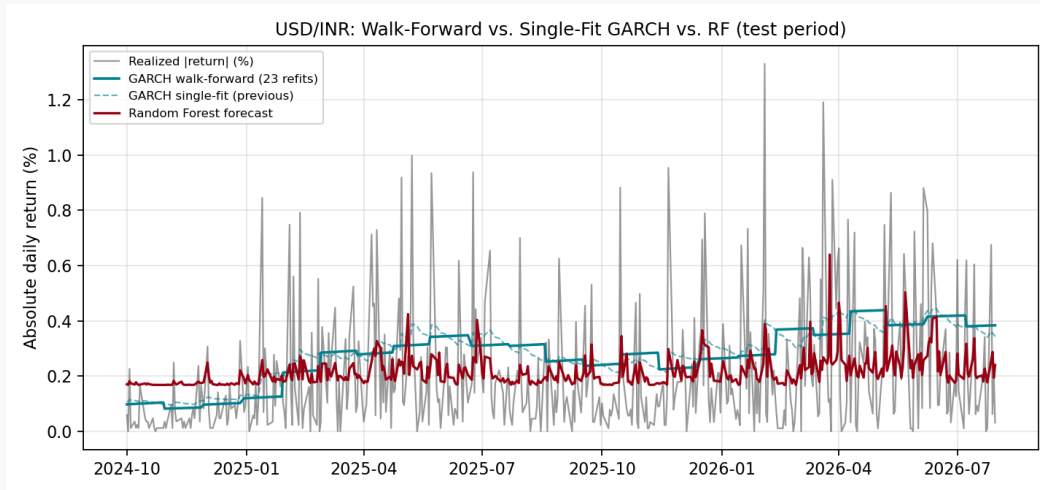
| Pair    | n  | R <sup>2</sup> | Sig. predictors?                       |
|---------|----|----------------|----------------------------------------|
| USD/INR | 89 | 5.0%           | No (best p = 0.17)                     |
| EUR/USD | 89 | 12.4%          | Yes — infl., trade bal., GDP (p < .04) |
| GBP/USD | 89 | 3.5%           | No (best p = 0.19)                     |
| USD/JPY | 89 | 3.7%           | No (best p = 0.33)                     |

## Interpretation

Expanding to the full 4-predictor specification (rate, inflation, trade balance, and GDP-growth differentials — restoring Japan's inflation term via World Bank annual CPI) raises R<sup>2</sup> for every pair. For EUR/USD, three of four coefficients are significant (inflation p=.023, trade balance p=.017, GDP growth p=.039) — the project's first significant RQ1 result. VIF stayed low (1.0-1.8) across all pairs — multicollinearity is not the cause.

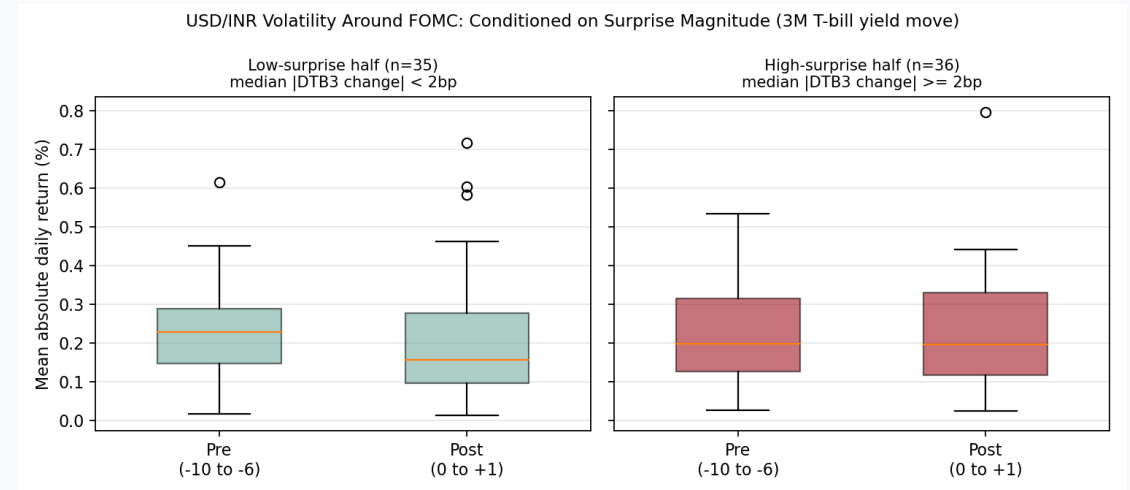
*Takeaway: the Meese-Rogoff puzzle still holds for USD/INR, GBP/USD, and USD/JPY — but EUR/USD's deep, liquid market shows real, statistically significant sensitivity to macro fundamentals at a monthly horizon.*

# Results: RQ2 Volatility & RQ4 Event Study



## RQ2 — GARCH (walk-forward) 0.231 | RF 0.227 | Naive 0.298

Re-estimated with true walk-forward refitting (23 refits, no look-ahead) rather than a single in-sample fit — both real models still beat the naive baseline by ~22-25%, and the GARCH-vs-RF gap remains non-significant (paired t,  $p = 0.78$ ).



## RQ4 — $n = 71$ dates, split by surprise size (T-bill proxy)

Splitting on  $|\text{3-month T-bill yield change}|$  (a Kuttner-style surprise proxy): the high-surprise half shows a +2.2% post-announcement volatility rise vs. -6.3% for the low-surprise half — the expected direction, but  $n=36/35$  is too small to reach significance.

**Cross-cutting takeaway: volatility is more forecastable than direction — and surprise size, not meeting occurrence, drives the (directional, not-yet-significant) reaction.**

# Implementation & User Benefit

## 1 Corporate Treasury

Time FX hedges around scheduled policy dates using event-study risk flags

## 2 FX Traders / PMs

Short-term volatility forecasts (GARCH/RF) as a tactical risk-sizing input

## 3 Import/Export Firms

Early-warning signal on elevated volatility risk before it hits margins

## Proposed Deliverable

- A dashboard-style early-warning summary built directly from model outputs — not a raw academic forecast.
- Flags elevated volatility risk ahead of scheduled central-bank announcements (RQ4) for the currency pairs a user holds exposure in.
- Surfaces the RQ1/RQ2 model outputs (direction tendency, volatility forecast) side by side so users can weigh macro-driven vs. statistical signals.
- Recommends model family by decision horizon: macro regression for medium-term hedging, GARCH/ML for short-term tactical calls.

## CONCLUSION

# Short-term FX direction resists prediction — but volatility itself is a tractable, actionable signal.

- RQ1, RQ2, and RQ4 mostly fail to reject their null hypotheses — a coherent, literature-consistent pattern — but the full 4-predictor RQ1 model breaks that pattern for EUR/USD ( $R^2=12.4\%$ , 3 of 4 coefficients significant), showing the null is not universal.
- For USD/INR, GBP/USD, and USD/JPY, RQ1 + RQ4 still confirm the Meese-Rogoff / Kuttner picture: fundamentals and scheduled events alone don't move short-term FX in a statistically detectable way.
- RQ2 shows real practical value under a genuine walk-forward test: both GARCH and Random Forest cut volatility-forecast error by ~22-25% versus a naive baseline — this is the actionable core of the project.
- The honest, rigorously-tested results — including the null ones — are themselves the deliverable expected of a capstone at this stage; they sharpen, rather than undermine, the recommendation ahead.

# Limitations & Future Work

## Limitations

- RQ3 (sentiment) is architecturally defined but not yet executed — GDELT collection remains.
- Eurozone has no native monthly trade-balance aggregate; it is built by summing 20 euro-area members' annual balances (Eurostat), forward-filled monthly.
- India/UK/Japan inflation now uses World Bank annual CPI (current through 2025, replacing FRED series stale since Mar 2025 / dead since Jun 2021) — real and current, but coarser than a monthly series.
- RQ4's surprise-magnitude split (3-month T-bill yield proxy, not true consensus data) shows the expected direction, but the  $n=36/35$  halves are too small to reach significance.
- Walk-forward GARCH refits every 20 trading days for tractability; a daily-refit cadence was not attempted due to compute cost.

## Future Work

- Complete GDELT sentiment pipeline (VADER + FinBERT) and run the RQ3 comparison.
- Acquire a true survey-based policy-surprise series (e.g., Bloomberg/Reuters consensus) to replace the T-bill-yield proxy used here, and re-run RQ4 with adequate power.
- Source a native monthly Eurozone trade-balance series (e.g., ECB SDW) to replace the member-state aggregation used here.
- Extend RQ2 to all four pairs and add XGBoost/LSTM to the comparison, as originally scoped.
- Deploy the recommended early-warning dashboard as a live internal tool for pilot users.

# Bibliography (APA 7)

Araci, D. (2019). FinBERT: Financial sentiment analysis with pre-trained language models. arXiv.

Bank for International Settlements. (2022). Triennial Central Bank Survey: OTC foreign exchange turnover in 2022.

Bender, E. M., & Friedman, B. (2018). Data statements for NLP. TACL, 6, 587-604.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. J. of Econometrics, 31(3), 307-327.

Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5-32.

Button, K. S. et al. (2013). Power failure: Why small sample size undermines neuroscience. Nat. Rev. Neurosci., 14(5), 365-376.

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. KDD 2016, 785-794.

Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. JBES, 13(3), 253-263.

Engle, R. F. (1982). Autoregressive conditional heteroscedasticity. Econometrica, 50(4), 987-1007.

Federal Reserve Bank of St. Louis. (n.d.). FRED economic data [Data set]. fred.stlouisfed.org

Geburu, T. et al. (2018). Datasheets for datasets. arXiv.

Gelman, A., & Hill, J. (2007). Data analysis using regression and multilevel/hierarchical models. Cambridge Univ. Press.

Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. Neural Computation, 9(8), 1735-1780.

Hutto, C., & Gilbert, E. (2014). VADER: A parsimonious rule-based sentiment model. ICWSM 2014, 216-225.

Kuttner, K. N. (2001). Monetary policy surprises and interest rates. J. of Monetary Economics, 47(3), 523-544.

Meese, R. A., & Rogoff, K. (1983). Empirical exchange rate models of the seventies. J. of Int'l Economics, 14(1-2), 3-24.

O'Brien, T., Sukumar, A., & Helfert, M. (2013). The value of good data - A quality perspective. ICEIS 2013, 555-562.

Rahm, E., & Do, H. H. (2000). Data cleaning: Problems and current approaches. IEEE Data Eng. Bulletin, 23(4), 3-13.

Tetlock, P. C. (2007). Giving content to investor sentiment. The Journal of Finance, 62(3), 1139-1168.

Torralba, A., & Efros, A. A. (2011). Unbiased look at dataset bias. CVPR 2011, 1521-1528.